Niranjan Murugarasu

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Career Objective

Final-year B.Tech Artificial Intelligence and Data Science student specializing in full-stack web development (Next.js, React, Node.js/Express, Spring Boot) and agentic AI systems (autonomous tool-calling assistants, LLM fine-tuning and evaluation), backed by a strong foundation in Data Structures & Algorithms (500+ LeetCode problems solved) and hands-on experience shipping scalable, production-style software with Dockerized deployment pipelines. Comfortable translating ambiguous problems into well-tested, working code in fast-moving, Agile team environments.

Education

B.Tech in Artificial Intelligence and Data Science — **2023–2027**
Dr. N.G.P Institute of Technology, Coimbatore — CGPA: 8.65

Higher Secondary — **2023**
Elgi Matric Hr. Sec. School, Coimbatore — 86%

Technical Skills

- **Programming & Databases:** Java, Python, SQL, PostgreSQL, NoSQL
- **Full-Stack Development:** React, Next.js, Node.js/Express, Spring Boot, REST APIs
- **Agentic AI & GenAI/LLMs:** Agentic AI Development (autonomous tool-calling assistants), Prompt Engineering, LLM Output Evaluation & Validation, RAG, Vector Databases, LoRA/QLoRA Fine-Tuning, Transformers
- **Core CS:** Data Structures & Algorithms, OOPs, DBMS, Operating Systems, System Design
- **Deployment & Tools:** FastAPI, Docker, CI/CD, Git/GitHub, Streamlit
- **ML / Data Science:** PyTorch, Scikit-learn, Supervised/Unsupervised ML, NLP, Feature Engineering
- **Mathematics:** Probability, Statistics, Linear Algebra, Optimization

Internship

Graphic Designer Intern, *Visaithalam Solutions* — October 2023 – February 2024
Delivered branding and marketing creatives across client projects under deadline constraints.

Projects

1) SwiftDesk AI – Intelligent Ticket Triage, SLA Escalation & Auto-Routing System — 2026

- Engineered an automated multi-tier support routing backend in Java 25 and Spring Boot 3, implementing a keyword-based NLP triage engine for instant category classification, severity resolution, and customer priority dishonesty (trap) detection.
- Designed a real-time load-balancing assignment engine across L1, L2, and L3 engineering tiers based on active workload capacity ratios (`active_tickets / max_capacity`), featuring automated priority upgrade triggers upon SLA escalation.
- Architected a persistent relational data model in PostgreSQL 17 via Spring Data JPA and HikariCP, capturing full lifecycle audit logs across `tickets`, `ticket_history`, and transactional `email_logs` with non-destructive database seeding.
- Built a dynamic React 19 and Material-UI customer/support dashboard connected via an Axios API service with automatic mock-fallback resilience for offline testing and seamless RESTful state synchronization.

2) PrepAI – AI-powered Interview Simulation Platform — 2025

- Designed and iterated prompt templates for Llama 3.3 70B (via Groq) to generate context-aware interview questions and adaptive follow-ups conditioned on job descriptions and prior candidate responses.

- Built a 5-metric evaluation framework (technical accuracy, communication, business impact, confidence, structured thinking) to score AI-generated candidate responses, translating qualitative LLM output into a structured 0–10 rubric with STAR-method feedback.
- Shipped the system on a full-stack architecture (Next.js, Node.js/Express) with Groq-hosted Llama 3.3 70B inference, generating detailed, actionable performance reports per candidate.

3) Neural Code Smell Detector – Hybrid AI + Rule-Based Code Quality Evaluation System 2025

- Fine-tuned Microsoft's CodeBERT (PyTorch) for multi-label classification of 4 Python code smells on 1.9K GitHub refactoring samples.
- Ran a stratified 80/20 held-out evaluation and diagnosed a severe class-imbalance failure mode invisible under aggregate accuracy: 99.5% F1 on the majority class (bad naming, 96% of labels) but 0% recall on the 3 minority smell classes – surfaced only via per-class macro-F1 analysis (24.9% macro F1 vs. 97.1% micro F1).
- Combined model predictions with AST- and Radon-based static analysis (cyclomatic complexity, nesting depth, naming quality) into a single, explainable 0-100 code-quality score.
- Built a feedback layer surfacing the specific factors behind each score alongside automated refactoring suggestions, shipped through an interactive Gradio UI.

4) AI-powered Stock Price Prediction System 2024

- Built end-to-end time-series forecasting using LSTM, GRU, and Transformer models.
- Implemented backtesting, cross-validation, and ensemble modeling.
- Deployed inference using FastAPI + Docker with monitoring.

5) MedRx – Fine-Tuned Clinical Drug & Treatment Recommendation LLM 2026

- Built a post-inference output-validation layer for a fine-tuned clinical LLM, running automated drug-allergen cross-reactivity and dosage range checks against a structured (Pydantic-parsed) schema to catch unsafe model outputs before they reach an end user.
- Designed a 7-gate data-quality filtering pipeline (PHI scrubbing, MD5 deduplication, drug-class-templated synthetic augmentation) that reduced a 172K-record raw corpus to 13.7K high-quality training samples (92% rejection rate) – directly defining what "high-quality" meant for the fine-tuning task.
- Fine-tuned Microsoft's Phi-4-mini-Instruct (3.8B) via 4-bit QLoRA (NF4) with Unsloth, training an 8.9M-parameter LoRA adapter (0.23% of base) on the resulting 13,728-sample clinical-reasoning dataset.
- Root-caused and resolved a Triton JIT compiler segfault specific to Ada Lovelace GPUs via native stack-trace analysis; published the resulting model publicly on Hugging Face.

Co-Curricular Activities

- Built **StalePR**, an agentic AI assistant integrating with Slack and GitHub to autonomously manage stale pull requests – developed for the Slack Agent Builder Challenge.
- Launched and author the monthly LinkedIn newsletter **The AI Edge**, sharing insights on artificial intelligence, emerging technologies, and industry trends.
- Active member of the Robotics Club, participating in technical activities and collaborative engineering projects.
- Presented a research paper on an AI-powered Healthcare Data Handling Platform at **Coimbatore Institute of Technology (CIT)**, Coimbatore.
- Presented a research paper on **Solar Technology** at **KPR Institute of Engineering and Technology**, Coimbatore.
- Participated in the **Illuminate Entrepreneurship Bootcamp** organized by the E-Cell, IIT Bombay, focusing on innovation, startup ideation, and entrepreneurship.

Certifications

- AI Fluency Framework – **Anthropic** 2025
- Generative AI Fundamentals – **Databricks** 2026
- Elements of AI – **University of Helsinki** 2026
- GenAI Powered Data Analytics Job Simulation – **Tata** 2026
- Human-Computer Interaction – **NPTEL** 2026